

DRAWING LIST FOR TEST PACKAGE

PROJECT NO.: 22328D

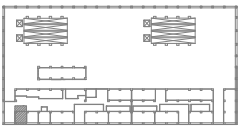
PROJECT TITLE : M15X Ph2 Project

NaOCI-565

No	PKG No	UNIT	LINE NUMBER			P&ID		REMARK
			ISO DRAWING No	SHEET	REV	P&ID No	REV	
1	565	4FA	OD25-NaOCI-PV3-N-6509-01	1 OF 1	0	CJ-M15W13-PR-A20-000057	2	
2	565	4FA	OD32-NaOCI-PV3-N-6511-01	1 OF 1	0	CJ-M15W13-PR-A20-000057	2	
3	565	4FA	OD32-NaOCI-PV3-N-6513-01	1 OF 3	0	CJ-M15W13-PR-A20-000057	2	
4	565	4FA	OD32-NaOCI-PV3-N-6513-02	2 OF 3	1	CJ-M15W13-PR-A20-000057	2	
5	565	4FA	OD32-NaOCI-PV3-N-6513-03	3 OF 3	0	CJ-M15W14-PR-A20-000116	1A	

NaOCl-565

대외비



KEY PLAN (4F)

NOTE

- DIAPHRAGM PUMP
MAT'L 표기 : HEAD / DIAPHRAGM / CHECK BALL
- 약품공급에 따라 LINE 색상을 구분하여야 한다.
- CHEMICAL STORAGE TANK & PUMP CURB에서 발생하는 DRAIN에 인근의 TRENCH로 침수되어, DRAIN HEADER로 이용된다.
- PUMP VENDOR 규격.
- CHEMICAL UNLOADING STATION에서 LEVEL을 볼 수 있도록 한다.
- TANK의 역면 레벨 게이지는 VENDOR에서 공급함.
- INTERLOCK NO. I-4657
(W13-CTP-NaOCl-4677 AUTO STOP)
: REMOTE STATUS에서 W13-T-NaOCl-4609 LEVEL (U-W6019) HIGH HIGH STOP.
- INTERLOCK NO. I-4624
(W13-CDP-NaOCl-4660 A/B, 4661 A/B AUTO STOP)
: REMOTE STATUS에서 W13-T-NaOCl-4609 LEVEL (U-W6020) LOW LOW STOP.
CIF MBR CIP/BACK WASH 시 START/STOP.
- DELETED.
- DELETED.
- DELETED.
- FOR PH4 CONNECTION NOZZLE.
- REUSE SYSTEM 및 BRINE 처리 SYSTEM의 공급용 PUMP 수단에 설치함.
- OPERATING CAPACITY : 1 LPM
OPERATING PRESSURE : 0.39 MPa
- OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.39 MPa
- INTERLOCK NO. I-4625
(W14-CDP-NaOCl-4660/4661, W13-CDP-NaOCl-4660 B/4661 B AUTO STOP)
: REMOTE STATUS에서 W13-T-NaOCl-4609 LEVEL (U-W6020) LOW LOW STOP.
CIF MBR CIP/BACK WASH 시 START STOP.
- OPERATING CAPACITY : 1 LPM
OPERATING PRESSURE : 0.39 MPa
- OPERATING CAPACITY : 10 LPM
OPERATING PRESSURE : 0.39 MPa

2	REVISED AFC	24.06.21
1	REVISED AFC	23.02.28
0	APPROVAL FOR CONSTRUCTION	22.09.23
수정 REV.	설명 DESCRIPTION	날짜 DATE

제 도	한 진 희
DRAWN BY	
검 토	심 문 용
CHECKED BY	
승 인	박 동 영
APPROVED BY	
제 작 사	

SK 에코플랜트

공 사 명
PROJECT TITLE
M15X Ph3 WWT Project

도면명
DWG. TITLE
PIPING & INSTRUMENT DIAGRAM
NaOCl STORAGE TANK
& FEED PUMP

속 척
SCALE
NONE

날 짜
DATE
2024. 06. 21

PROJECT NO. 22330D

도면번호 SHEET NO. CJ-M15W13-PR-A20-000057

2

ITEM NO.	W13-T-NaOCl-4609
SERVICE	NaOCl STORAGE TANK
Q'TY	1
TYPE	VERTICAL & CYLINDRICAL
SIZE	ID x H (EH 1.6)
CAPA.	m3 7(EH 6)
MAT'L	STS304 + FRP

SIGNAL

M15W13-PR-A20-000013

FROM CIF MBR LCP

SIGNAL

M15W14-PR-A20-000113

FROM CIF MBR LCP

SIGNAL

M15W14-PR-A20-000315

TO W14-CDP-NaOCl-4703 A/B,

W14-CDP-NaOCl-4704 A/B,

W14-CDP-NaOCl-4705 A/B,

W14-CDP-NaOCl-4752 A/B,

W14-CDP-NaOCl-4753 A/B

NaOCl

M15W14-PR-A20-000315

TO REUSE SYSTEM

NaOCl FEED PUMP

SUCTION HEADER

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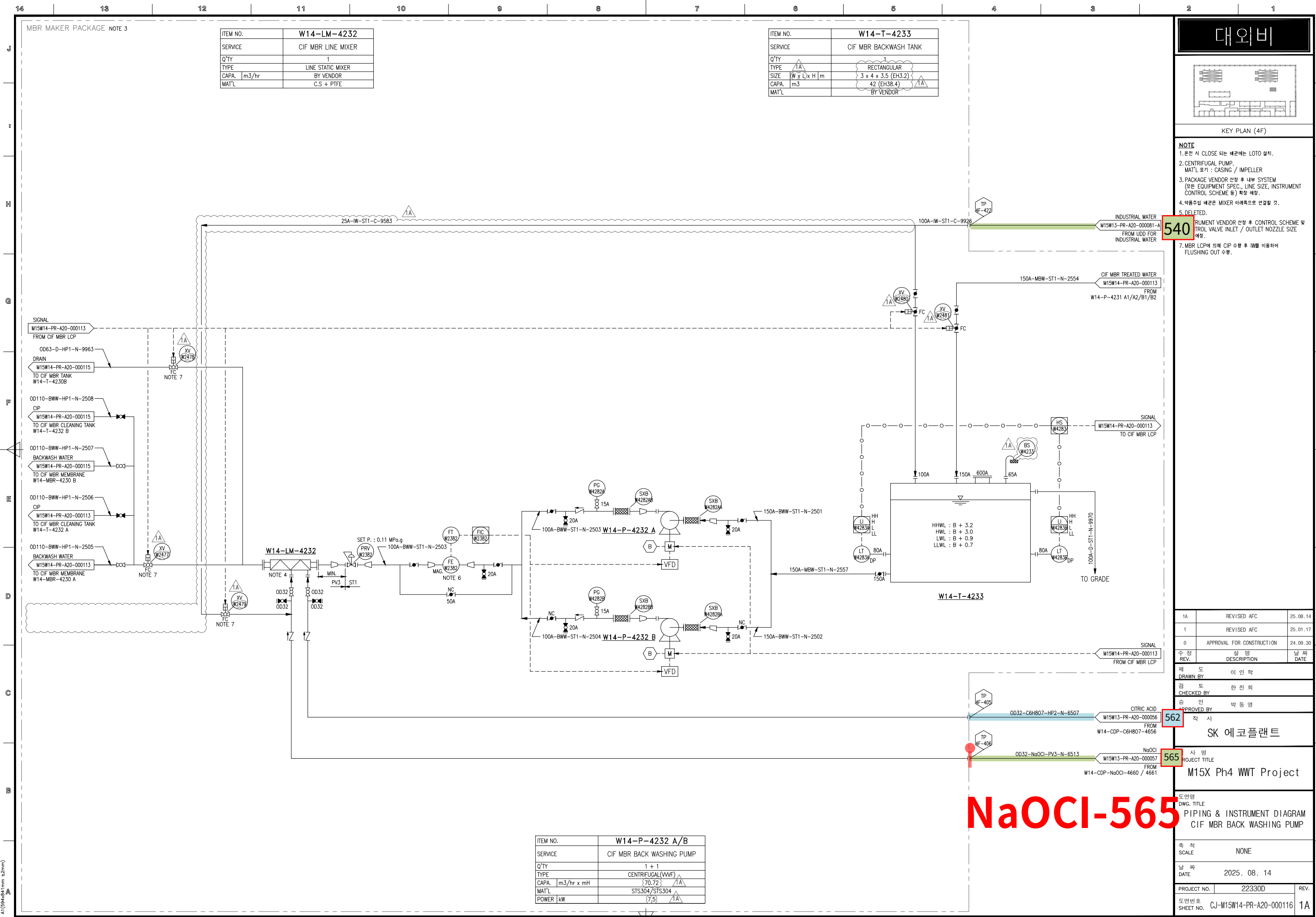
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ITEM NO.	W14-LM-4232
SERVICE	CIF MBR LINE MIXER
Q'TY	1
TYPE	LINE STATIC MIXER
CAPA.	m3/hr
MAT'L	C.S + PTFE

ITEM NO.	W14-T-4233
SERVICE	CIF MBR BACKWASH TANK
Q'TY	1
TYPE	RECTANGULAR
SIZE	3 x 4 x 3.5 (EH3.2)
CAPA.	m3
MAT'L	42 (EH38.4)
	BY VENDOR

ITEM NO.	W14-P-4232 A/B
SERVICE	CIF MBR BACK WASHING PUMP
Q'TY	1 + 1
TYPE	CENTRIFUGAL(VWF)
CAPA.	m3/hr x mH
	{70.72} / 1A
MAT'L	STS304/STS304
POWER	kW
	{7.5} / 1A

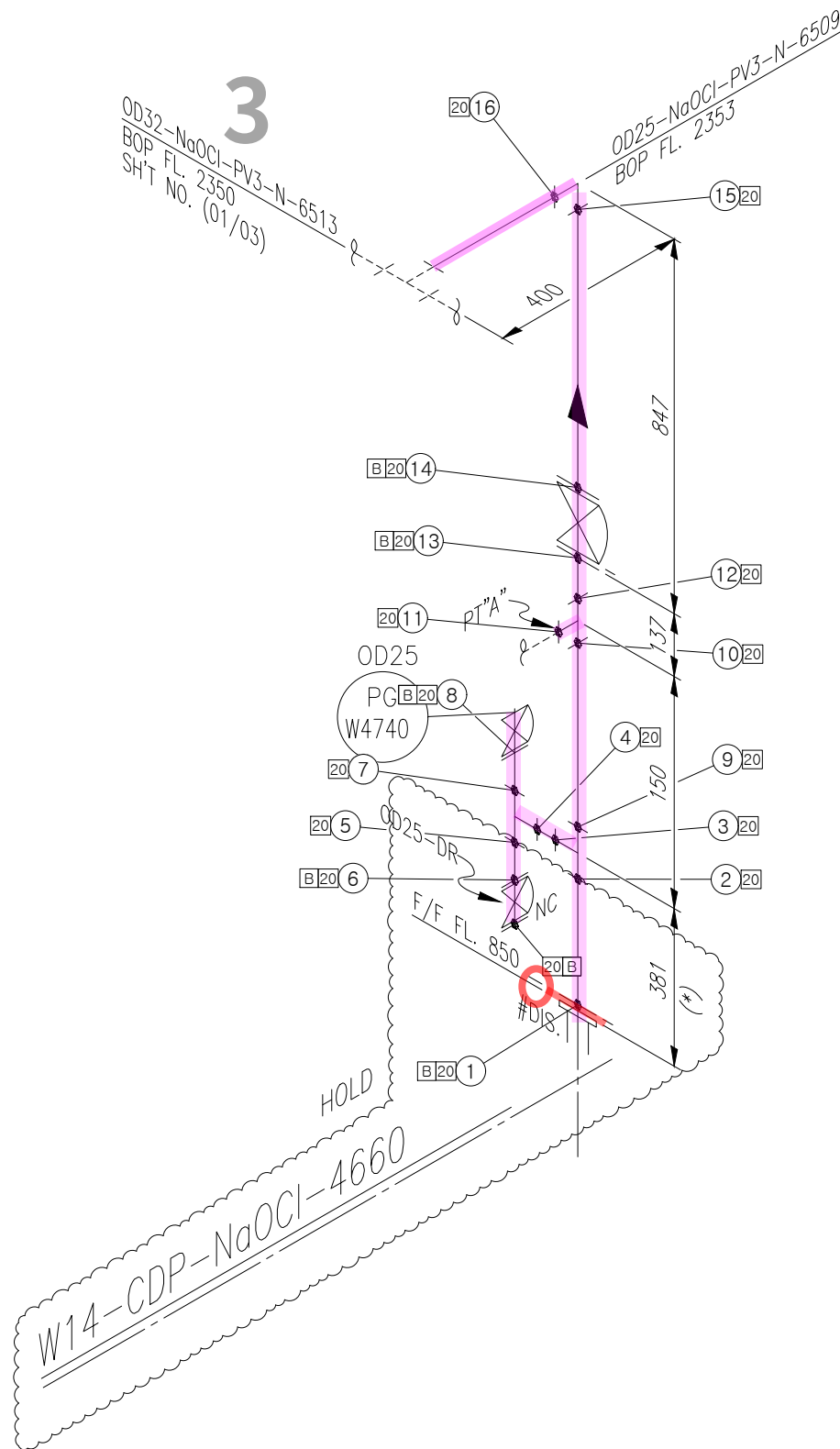
대외비

KEY PLAN (4F)

NOTE

1. 운전 시 CLOSE 되는 배관에는 LOTO 설치.
2. CENTRIFUGAL PUMP.
MAT'L 표기 : CASING / IMPELLER
3. PACKAGE VENDOR 선정 후 내부 SYSTEM
(온도 EQUIPMENT SPEC., LINE SIZE, INSTRUMENT
CONTROL SCHEME 등) 확정 예정.
4. 약품주입 배관은 MIXER 아래쪽으로 연결할 것.
5. DELETED.
6. PACKAGE VENDOR 선정 후 CONTROL SCHEME 및
CONTROL VALVE INLET / OUTLET NOZZLE SIZE
예정.
7. MBR LCP에 의해 CIP 수행 후 IW를 이용하여
FLUSHING OUT 수행.

1A	REVISED AFC	25.08.14
1	REVISED AFC	25.01.17
0	APPROVAL FOR CONSTRUCTION	24.09.30
수정 REV.	실명 DESCRIPTION	날짜 DATE
제 DRAWN BY	이인학	
검 CHECKED BY	한진희	
승 APPROVED BY	박동명	
작 작사		
SK 에코플랜트		
사명 PROJECT TITLE	M15X Ph4 WWT Project	
도면명 DWG. TITLE	PIPING & INSTRUMENT DIAGRAM CIF MBR BACK WASHING PUMP	
축척 SCALE	NONE	
날짜 DATE	2025. 08. 14	
PROJECT NO.	22330D	REV.
도면번호 SHEET NO.	CJ-M15W14-PR-A20-000116	1A

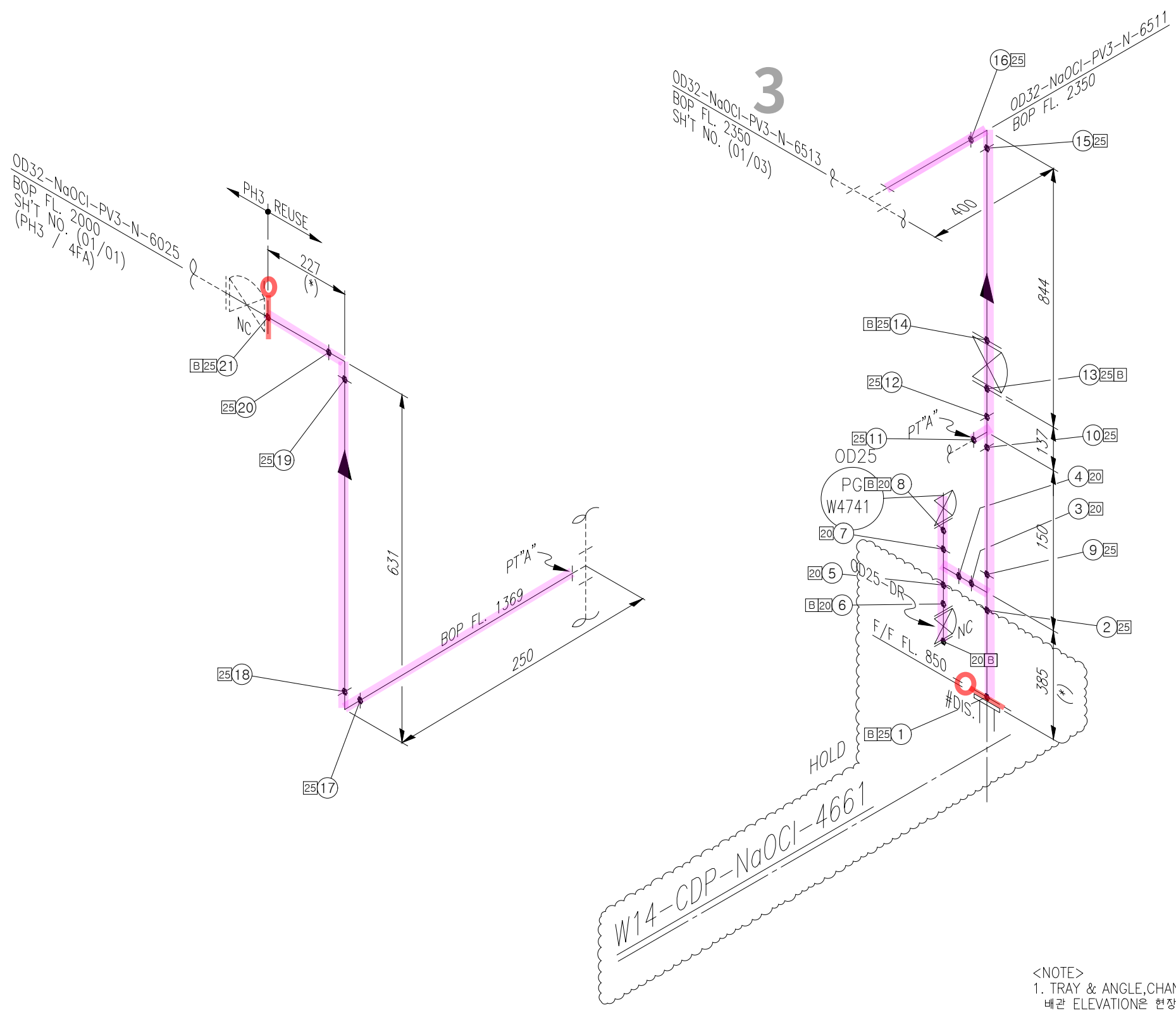


1. TRAY & ANGLE, CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

[illegible]

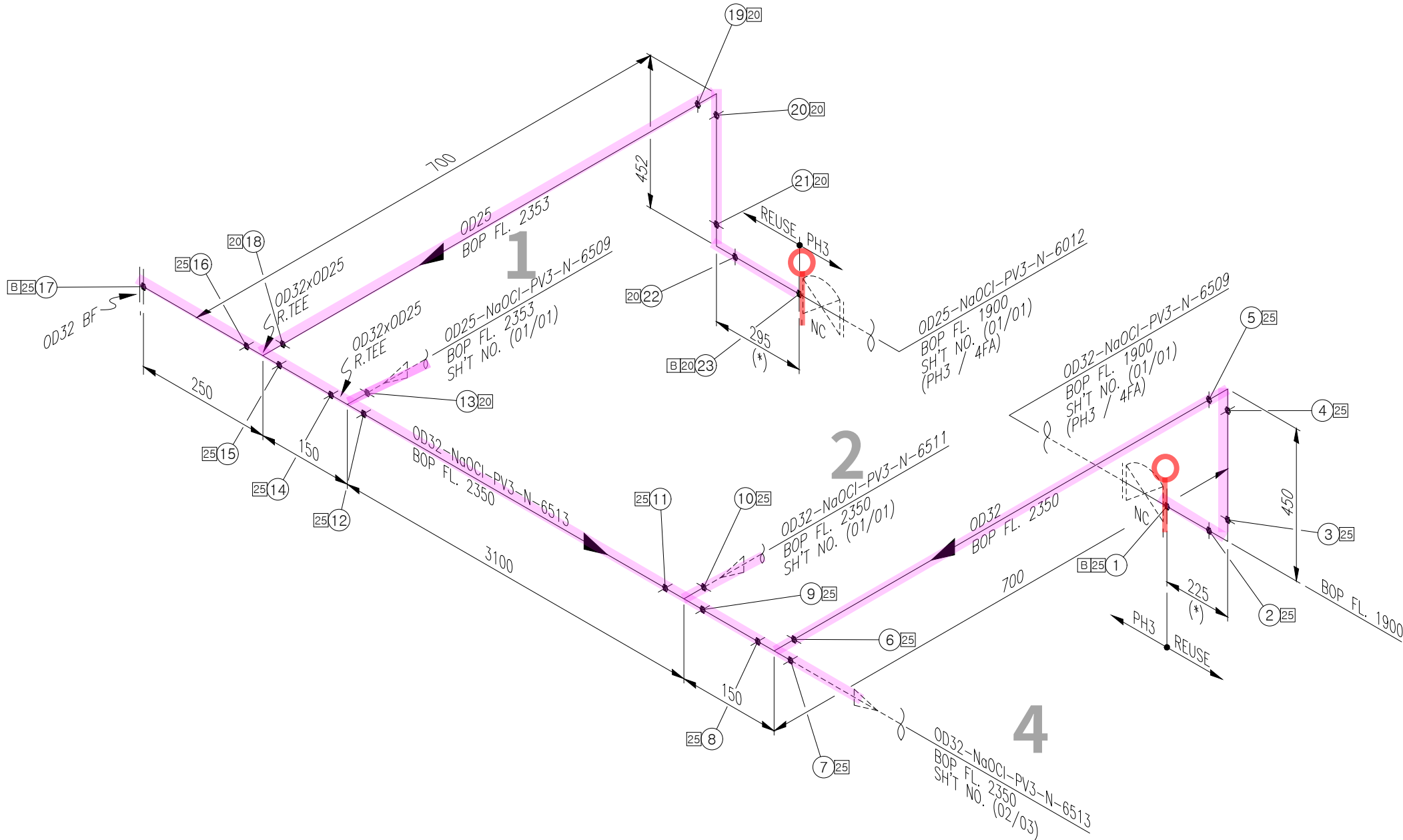
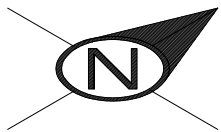
CLIENT		1	
		PROJECT NAME	
		M15X Ph4 Project	
COMMODITY NaOCI		JOB NO.	
		22330D	
LINE NO.	0D25-NaOCI-PV3-N-6509	SH'T NO.	REV.
		01 / 01	0

* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1 * STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7 * GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH * STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1													
INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14						
	THICK.	-	DESIGN	4.4	60	UNIT	4FA						
			OPERATING	4.0	AMB	REF. PLAN DWG.							
						CJ-M15W14-PI-A30-040009-A							
						REF. P & ID		0	25.01.24	APPROVED FOR CONSTRUCTION	황재중	안남후	박동영
						CJ-M15W14-PR-A20-000057		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'



CLIENT  2

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	4.4	60	UNIT	4FA								M15X Ph4 Project	
			OPERATING	4.0	AMB	REF. PLAN DWG.								COMMODITY	JOB NO.	
						CJ-M15W14-PI-A30-040009-A			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOCl	22330D	
						REF. P & ID		REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	SH'T NO.	REV.
						CJ-M15W14-PR-A20-000157								0D32-NaOCl-PV3-N-6511	01 / 01	0



<NOTE>
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배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST



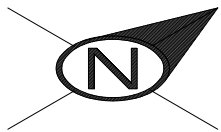
ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD32	5.0M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
P	OD25	1.5M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
E90	OD32	2EA	ELBOW 90 DEG, CPVC, PE, SOCKET FUSION, SDR13.5
E90	OD32	2EA	ELBOW 90 DEG, CPVC, PE, SOCKET FUSION, SDR13.5
T	OD32	2EA	TEE EQUAL, CPVC, PE, SOCKET FUSION, SDR13.5
TR	OD32xOD25	2EA	TEE RED, CPVC, PE, SOCKET FUSION, SDR13.5
F	OD32	2EA	FLG, KS 10K, FF, STUB FLG CPVC / BACKING RING(PP)+INSERT(STEEL), SDR13.5
F	OD25	1EA	FLG, KS 10K, FF, STUB FLG CPVC / BACKING RING(PP)+INSERT(STEEL), SDR13.5
FB	OD32	1EA	FLG BLIND, KS 10K, FF, CPVC, SDR13.5
G	OD32	2EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
G	OD25	1EA	GASKET, KS 10K, FF, PTFE+EPDM, 3.0MM TH'K
B	OD32	2SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
B	OD25	1SET	MACHINE BOLT & NYLON NUT :STS 304 / W WASHERS
SH	OD32	2EA	SHIELD
SH	OD25	1EA	SHIELD



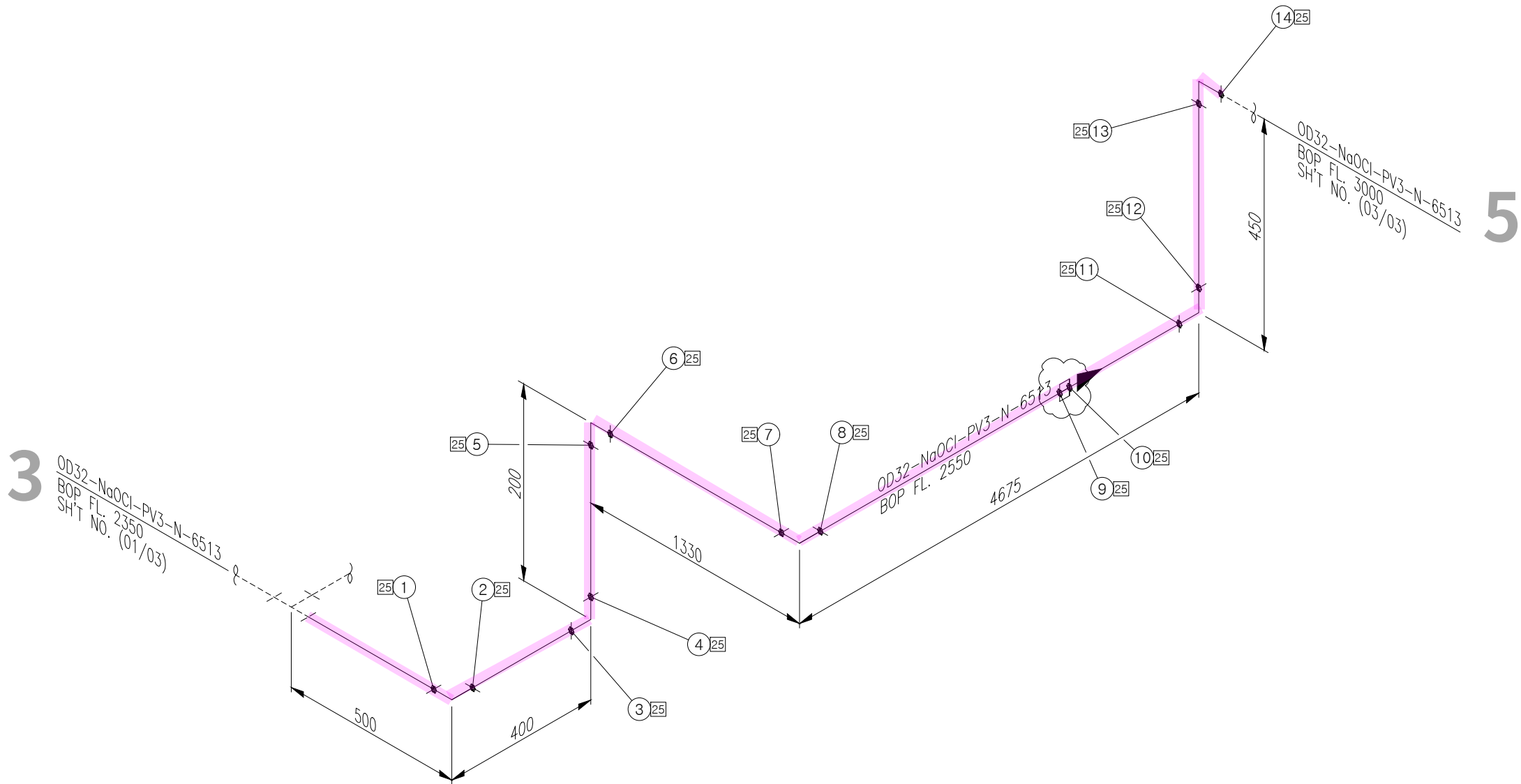
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* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PTFE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14										CONTRACTOR	PROJECT NAME	
	THICK.	—	DESIGN	4.4	60	UNIT	4FA											M15X Ph4 Project	
			OPERATING	4.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040009-A											COMMODITY	JOB NO.	
						REF. P & ID CJ-M15W14-PR-A20-000157			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영				LINE NO.	SH'T NO.	
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D				0D32-NaOCl-PV3-N-6513	01 / 03	REV. 0



ITEM	SIZE	Q'TY	DESCRIPTION
PV3			
P	OD32	7.5M	PIPE, CPVC, PE, SOCKET FUSION, SDR13.5
E90	OD32	6EA	ELBOW 90 DEG, CPVC, PE, SOCKET FUSION, SDR13.5
SC	OD32	1EA	SOCKET, CPVC, PE, SOCKET FUSION, SDR13.5



- <NOTE>
1. TRAY & ANGLE,CHANNEL SUPPORT가 적용되는 배관 ELEVATION은 현장 맞춤으로 한다.
2. (*) FIELD ADJUST

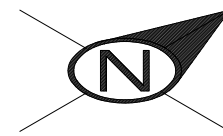
* PIPE : P/P1 * 90° ELBOW : E90 * 45° ELBOW : E45 * TEE : T * RED.TEE : TR * STUB-IN : STU * REINFORCING PAD : RP * CON.RED : RC * ECC.RED : RE * CON.SWAGE : RCS/RCS1 * ECC.SWAGE : RES/RES1 * NIPPLE : NP/NP1/NP2 * CAP : CAP/CAP1 * PLUG : PLG * FULL COUPLING : FC * HALF COUPLING : HF/HF1
* STUB-IN : STU * SOCKET(HDPE/PVC) : SC * SADDLE(HDPE) : SD * FLANGE : F/F1 * GASKET : G/G1/G2 * BOLT & NUT : B/B1 * BLIND FLANGE : FB/FB1 * GATE VA. : VG/VG1/VG2 * BALL VA. : VB/VBLO/VB1/VB1LO/VB2/VB2LO * BUTTERFLY VA. : VF/VF4 * CHECK VA. : VC/VC1/VC2/VC3/VC4/VC5/VC6/VC7
* GLOBE VA. : VL/VL1/VL2 * DIAPHRAGM VA. : VD/VDLO * DAMPER : VP * QUICK COUPLING : SQC * FLEXIBLE JOINT(STEEL) : FLXB * FLEXIBLE JOINT(RUBBER) : FLXR * FLEXIBLE JOINT(STEEL PIPE LINED) : FLXL * FLEXIBLE HOSE : FLXH * STEAM TRAP : ST * SPRAY NOZZLE : SPN * EMERGENCY SHOWER : SSH
* STRAINER(Y-TYPE) : YSTR * STRAINER(T-TYPE) : TSTR * STRAINER(BUCKET-TYPE) : BSTR * ANCHOR PLATE : STOP * BIRD SCREEN : BS * ISOLATION KIT : IK * UNLOADING BOX : UB * AIR TRAP : AT * LOTO : LOTO * SIGHT GLASS : SG * PRESSURE GAUGE : PG * SHILED : SH * SPECIAL BOLT & NUT : BL/BL1

CLIENT



4

INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA	W14							CONTRACTOR	PROJECT NAME		
	THICK.	—	DESIGN	4.4	60	UNIT	4FA								M15X Ph4 Project		
			OPERATING	4.0	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040009-A			25.03.31	REVISED AS MARKED	황래중	안남후	박동영	COMMODITY	JOB NO.		
						REF. P & ID CJ-M15W14-PR-A20-000157			25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동영	NaOCl	22330D		
								REV. NO.	DATE	DESCRIPTION	BY	CHK	APP'D	LINE NO.	OD32-NaOCl-PV3-N-6513	SH'T NO. 02 / 03	REV. 1



CLIENT		5
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INSULATION	CODE	N	CONDITION	PRESS. (kg/cm2g)	TEMP. (°C)	AREA W14						
	THICK.	—	DESIGN	1	60	UNIT 4FA						
			OPERATING	ATM	AMB	REF. PLAN DWG. CJ-M15W14-PI-A30-040007-A						
						REF. P & ID CJ-M15W13-PR-A20-000057		25.01.24	APPROVED FOR CONSTRUCTION	황래중	안남후	박동
							REV. NO.	DATE	DESCRIPTION	BY	CHK	APP

경고 : 본 도면은 SK HYNIX의 기업비밀 자료로서, 어떠한 경우에도 목적 외 사용, 복제, 제3자에게 공개, 누설 및 유출할 수 없으며, 회사는 유포자의 불법 사용으로 피해가 발생할 경우 유포자에게 민/형사상 책임을 요구할 수 있음